KISHORE MUNIYASAMY

Chennai, India | kishoremuniyasamy4@gmail.com | LinkedIn | github.com/KishoreXO | Portfolio

OBJECTIVE

Computer Science undergraduate specializing in AI & ML, proficient in Python, who builds AI and data-driven software end to end: statistical models and simulation, live data pipelines, real-time dashboards, and tested, documented, open-source releases. Seeking internships, research and project opportunities in AI and ML, and working toward training my own models.

EDUCATION

B.E., Computer Science & Engineering (AI & ML) 2025 – 2029
Sathyabama Institute of Science and Technology, Chennai, India

PROJECTS

Hammertime — Live Formula 1 Strategy Desk (GitHub) Open source, MIT
Python, FastAPI, React, Vite, TypeScript, Bayesian modelling, Monte Carlo simulation, OpenF1 / FastF1

- Built a live Formula 1 strategy dashboard that recommends when to pit, whether a pit window is open, or when to stay out, with tyre choice and rationale.
- Developed a strategy engine combining fuel-corrected lap times, per-driver Bayesian tyre-degradation models, track-status pit loss, traffic-aware rejoin estimates and undercut calculations.
- Implemented Monte Carlo race-plan simulation with uncertainty ranges, plus AI-assisted race-control, weather-risk and strategy judgements with an engine-only fallback.
- Created a multi-view dashboard for strategy calls, live track position, timing, race traces, whole-field decisions and pre-race tyre plans; supports live sessions, historical replays and a simulated demo race.
- Added backtesting, calibration and scorecard evaluation; demo-race results report 87% of plans within 1 second of the true optimum and a mean loss of 0.49 seconds.

Kestrel — Self-Hosted AI Job-Matching Platform (GitHub) Open source, MIT
Python, uv, SQLite, Node.js web dashboard, Docker, GitHub Actions, optional LLM APIs

- Built a privacy-first app that collects openings from official job APIs (Greenhouse, Lever, Ashby, USAJOBS, Adzuna) plus custom RSS/Atom/JSON feeds, and surfaces only roles that genuinely fit the user's profile.
- Designed a decision-model filtering pipeline: a free local keyword prefilter, then decision models that score every job on yes/no checks (experience, qualifications, location, work authorization, legitimacy) and weighted fit rubrics, with an optional LLM pass for nuanced calls; jobs are shown only at a score of at least 0.85 with 85% confidence, with an explanation and skill-gap view.
- Added preference learning from thumbs-up/down feedback, an application tracker (Saved, Applied, Interviewing, Offer), and caching that reclassifies a job only when the posting or profile changes.
- Made the optional LLM layer model-agnostic: works with hosted APIs or local models (Ollama, LM Studio, llama.cpp, vLLM) through any OpenAI-compatible endpoint; API keys never leave the user's machine.
- Built grounded answer drafting that uses only facts from the user's profile and never auto-applies; enforced responsible scraping with robots.txt compliance, rate limiting, a site denylist and a full audit log.

Mint — Autonomous Minecraft Survival Agent (GitHub) Open source, MIT
JavaScript, Node.js, Mineflayer / PrismarineJS, ESLint

- Built an autonomous agent that plays Minecraft survival on its own, from empty hands to a full diamond kit, across eight progression phases.
- Designed a priority-based behavior scheduler with a per-tick reflex layer for threats such as lava, drowning, hostile mobs and falls, plus per-world memory of stations, ores, deaths and hazards.
- Integrated an AI decision model for judgement calls (e.g. fight or flee), queried ahead of time so the agent never stalls, with a rule-based fallback when no model is available.
- Developed a live web dashboard with a 3D world view, behavior “brain board”, decision stream, death replay, radar and cross-run history.
- Wrote 35 automated test suites (including swimming tests on the real physics engine) and enforced code quality with ESLint.

TECHNICAL SKILLS

Languages	Python (proficient), JavaScript and TypeScript (working knowledge)
AI / ML	Bayesian modelling, Monte Carlo simulation, decision models, LLM integration, autonomous agents, local models (Ollama, LM Studio)
Tools	NumPy, FastAPI, React, Vite, Node.js, Mineflayer, Docker, Git, GitHub Actions, uv, SQLite, Playwright, ESLint
Practices	Automated testing, backtesting and evaluation, real-time dashboards, privacy by default, responsible data collection, open-source documentation

RESEARCH INTERESTS

Artificial Intelligence and Machine Learning, probabilistic modelling and simulation, real-time data systems, autonomous agents and decision-making, LLM applications, training custom models

LANGUAGES

English (fluent, strong spoken communication skills)